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Research on apartment energy platforms with shared solar and batteries

Advances and Challenges in Apartment Energy Platforms with Shared Solar and Batteries: Dashboards, Forecasting, NILM, Explainable AI, and Fair Energy Allocation (2019–2026)

Quick Reference

Key Findings Table

Theme	Approach	Key Findings	Benefits	Fairness/Privacy/Trust	Gaps/Novelty	Supporting Citations
Solar vs Grid Dashboards	Centralized, hybrid, P2P, blockchain	Centralized/shared systems outperform decentralized; dashboards with real-time, gamified, visual elements improve engagement; battery scheduling increases self-consumption	Up to 39% energy savings, 60% self-sufficiency, reduced peak demand	P2P models, blockchain, dynamic pricing enhance fairness; user-centered design builds trust	Need for standardized engagement metrics, cross-cultural studies	1 2 3 4 5 6 7 8 9 10 11
Solar/Load Forecasting	Ensemble, Bayesian, meta-learning, hybrid	Ensemble and Bayesian models improve uncertainty quantification; meta-learning enhances interpretability; prediction interval accuracy critical for battery scheduling	Robust forecasts, improved battery scheduling, reduced grid demand	Explainable AI increases transparency; federated learning preserves privacy	Lack of direct comparative studies on ensemble vs Bayesian for calibrated intervals	12 13 14 15 16 17 18 19
Appliance Monitoring (NILM)	Deep learning, federated, high-frequency, semi/transfer learning	High-frequency data and deep learning improve accuracy; federated learning preserves privacy; semi-supervised/transfer learning effective with limited labels	Accurate appliance recognition, privacy-preserving, scalable	Federated learning, edge processing, hybrid approaches mitigate privacy/trust issues	Need for standardized datasets, real-time frameworks	20 21 22 23 24 25 26 27 28
Explainable AI Load	SHAP, LIME, Grad-CAM,	XAI methods improve	Transparent load	User-tailored explanations,	Curse of dimensionality,	29 30 31 32 33 34

Theme	Approach	Key Findings	Benefits	Fairness/Privacy/Trust	Gaps/Novelty	Supporting Citations
Advice	clustering, hybrid	interpretability, user trust, actionable advice; clustering reduces computational cost; user-centric design enhances adoption	advice, improved decision-making, trust	workflow integration, contestability mechanisms	lack of standardized robustness testing	35
Fair Slot-Based Allocation	MDP, game theory, quota/reputation, multi-objective	Shapley value, Nash Bargaining, quota/reputation systems ensure fairness; multi-objective optimization balances economic and equity goals	Reduced disparities, equitable allocation, improved resilience	Reputation/quota mechanisms prevent exploitation; adaptive policies for building typologies	Lack of comparative studies, need for objective proximity analysis	36 37 38 39 40 41 42

Direct Answer

The research from 2019 to 2026 on apartment energy platforms integrating shared solar and battery storage demonstrates that centralized and hybrid configurations, advanced forecasting (ensemble/Bayesian/meta-learning), privacy-preserving NILM (federated/high-frequency/semi-supervised), explainable AI (SHAP, Grad-CAM, clustering), and fair slot-based allocation (game theory, quota/reputation) collectively improve operational efficiency, user engagement, fairness, privacy, and trust. Novel contributions include uncertainty-aware optimization for load scheduling and integrated fairness frameworks balancing economic and equity objectives amid operational uncertainties. For example, centralized solar dashboards have shown up to 39% energy savings and improved self-consumption 1 9, ensemble and Bayesian forecasting models capture uncertainties for reliable load predictions 43 44, federated NILM advances privacy-preserving architectures 23, and explainable AI approaches support real-time, user-centered load advice 29 32. Fair energy allocation strategies using the Shapley value and multi-objective optimization have demonstrated enhanced fairness and reduced disparities during outages 37 38.

Study Scope

- **Time Period:** 2019–2026
- **Disciplines:** Energy engineering, computer science (AI/ML), behavioral science, economics, urban planning
- **Methods:** Empirical studies, simulation, optimization (MDP, game theory), ensemble/Bayesian/meta-learning, federated learning, explainable AI, user-centered design, techno-economic analysis

Assumptions & Limitations

- **Assumptions:** Studies focus on apartment platforms with shared solar and batteries; only peer-reviewed journal articles (no conference papers); findings are generalizable across diverse apartment contexts.

- **Limitations:** Lack of standardized datasets for NILM; limited direct comparative studies on forecasting models; gaps in real-time XAI and federated NILM frameworks; economic viability sensitive to local market conditions and subsidies.

Suggested Further Research

- Comparative studies between dynamic ensemble and Bayesian deep learning for calibrated prediction intervals
- Standardized, reproducible frameworks for real-time NILM deployment in privacy-preserving federated settings
- Integration of emerging AI architectures (e.g., Transformers) within XAI for multivariate time series
- Blockchain-integrated federated learning for secure, verifiable energy transactions
- Objective proximity analysis for allocation models tailored to building typologies and resident profiles

1. Introduction

Background and Scope

Recent journal research (2019–2026) on apartment energy platforms integrating shared solar and batteries addresses the urgent need for sustainable, resilient, and equitable energy management in urban multi-unit dwellings. The main themes include:

- **Solar vs grid dashboards:** Real-time monitoring, user engagement, and economic optimization
- **Solar/load forecasting with prediction intervals:** Robust uncertainty quantification for operational decisions
- **Appliance monitoring (NILM):** Privacy-preserving, scalable appliance-level energy disaggregation
- **Explainable AI load advice:** Transparent, actionable recommendations for users and operators
- **Fair slot-based energy allocation during outages:** Equitable distribution of limited resources, tailored to building typologies and resident profiles

Motivation stems from the growing adoption of shared renewable resources, the need for user engagement, and the imperative to address fairness, privacy, and trust in energy management systems [1] [2] [36].

2. Theoretical Frameworks

Integration and Optimization of Shared Renewable Resources

- **Centralized vs decentralized architectures:** Centralized shared solar PV and battery systems outperform decentralized unit-level systems in energy utilization and inclusiveness [45].
- **Hybrid renewable configurations:** Combining PV, battery, hydrogen storage, and blockchain-based P2P trading enhances resilience and user empowerment [46].
- **Dynamic optimization:** MDP, game theory, and reinforcement learning optimize battery scheduling and energy allocation, balancing operational efficiency and cost-effectiveness [36] [43].

Uncertainty Quantification and Predictive Forecasting

- **Ensemble learning:** Cluster-based stacking, dynamic multi-model ensembles, and hybrid neural-SVM models improve prediction interval reliability [12] [15] [47].
- **Bayesian models:** BSTS and Bayesian deep neural networks provide robust uncertainty quantification and interpretability [13] [43].
- **Meta-learning:** Integrates diverse base models and temporal features, enhancing both accuracy and interpretability [14].

Explainability and User-Centric Interface Design

- **XAI techniques:** SHAP, LIME, Grad-CAM, clustering-based XAI, and hybrid approaches support transparent load advice and actionable insights [29] [30] [31].
- **User-centered design:** Visual elements (gauges, pie charts), gamification, and adaptive explanations improve engagement and trust [5] [6] [32].

Fairness, Equity, and Energy Allocation during Outages

- **Game theory:** Shapley value, Nash Bargaining, and envy-freeness ensure equitable allocation of shared resources [37] [38] .
- **Quota/reputation systems:** Prevent exploitation and promote fair utilization in community energy storage [48] [49].
- **Multi-objective optimization:** Balances economic benefits and fairness, adapting to building typologies and resident profiles [39] [40].

Privacy-Preserving and Federated Learning Methodologies

- **Federated learning:** Horizontal, vertical, and transfer learning architectures enable collaborative model training without sharing sensitive data [23] [50].
- **Edge processing:** Reduces privacy risks and implementation costs, supporting real-time NILM deployment [26] [28].

3. Methods & Data Transparency

- **Empirical studies:** Real-world deployments, user surveys, and behavioral experiments
- **Simulation and optimization:** MDP, game theory, stochastic and robust optimization, genetic algorithms
- **Machine learning:** Ensemble, Bayesian, deep learning, meta-learning, semi-supervised, transfer learning
- **Federated learning:** Distributed model training, privacy-preserving protocols, blockchain integration
- **Explainable AI:** SHAP, LIME, Grad-CAM, clustering, hybrid frameworks
- **User-centered design:** Usability testing, persona development, iterative feedback

Data transparency is achieved through open datasets (where available), standardized evaluation metrics, and reproducible frameworks, though gaps remain in NILM and real-time forecasting datasets [51] [52].

4. Critical Analysis of Findings

Solar vs Grid Dashboards

Design Approaches and Functionalities

- Centralized/shared systems achieve higher self-consumption and energy access ratios, outperforming decentralized models [45].
- Dashboards integrate real-time monitoring, forecasting, and control, supported by IoT and AI for performance optimization [3] [4].
- Hybrid renewable systems with blockchain-based P2P trading reduce grid demand and CO2 emissions, enhancing resilience [46].

User Interface Design and Engagement

- Visual elements (gauges, pie charts, flashing lights) improve engagement, reduce mental workload, and foster trust [6].
- Gamification increases knowledge and behavior change, though effects may plateau without sustained engagement strategies [53] [54].
- User-centered frameworks integrating cognitive, behavioral, and social factors optimize usability and sustained energy-saving behavior [55].

Battery Operational Strategies and Grid Consumption

- Adaptive battery scheduling increases self-consumption and reduces grid demand; community-shared storage outperforms individual systems [7] [8].
- Dynamic pricing and demand response enhance battery lifetime and economic performance [56] [57].

Economic Trade-offs and Payback Periods

- Economic viability depends on battery costs, tariffs, and subsidies; payback periods vary by country and consumption group [9] [37] [58].
- Community-shared systems offer higher net present values but are sensitive to capital costs and policy incentives [59].

Fairness and Peer-to-Peer Energy Sharing

- P2P models with dynamic pricing, reputational rankings, and blockchain enhance fairness and equity [2] [10] [11].
- Flexible market designs are needed to accommodate diverse user engagement and preferences [60].

Solar and Load Forecasting with Prediction Intervals

Forecasting Methods and Uncertainty Quantification

- Ensemble and Bayesian models provide robust prediction intervals, supporting reliable battery scheduling [12] [13].
- Meta-learning frameworks improve interpretability and actionable insights [14].

Ensemble and Hybrid Models for Prediction Intervals

- Dynamic multi-model ensembles and hybrid neural-SVM models outperform single models in interval reliability [15] [47].
- Bayesian deep learning quantifies both aleatoric and epistemic uncertainties, improving calibration [61].

Impacts of Prediction Interval Accuracy on Energy Management

- Accurate prediction intervals reduce operational risks, improve battery scheduling, and lower costs [16] [17].
- Uncertainty-aware optimization frameworks enhance resilience and economic efficiency [62] [63].

Interpretability and Explainability in Forecasting

- XAI methods (SHAP, LIME, Grad-CAM) improve transparency and user trust in forecasting models [14] [18].
- Combining multiple XAI techniques enhances reliability and actionable insights [64].

Calibration Challenges and Solutions

- Calibration is challenged by high variability in individual loads; hybrid modeling and clustering improve reliability [19] [65].
- Lack of direct comparative studies between ensemble and Bayesian models for calibrated intervals is a research gap [66].

Appliance Monitoring via Non-Intrusive Load Monitoring (NILM)

Advancements and Challenges in NILM

- High-frequency harmonic data and deep learning models significantly improve appliance recognition accuracy [20] [22].
- Real-time deployment is limited by data scarcity, computational costs, and lack of universal frameworks [51] [67].

Federated Learning for Privacy-Preserving NILM

- Federated learning enables collaborative model training without sharing raw data, preserving privacy [23] [50].
- Privacy-aware federated reinforcement learning coordinates distributed energy storage and load management [68].

High-Frequency Harmonic Signature Datasets

- Synthetic high-frequency datasets facilitate model selection and generalization testing, but require more storage and computation [69] [70].

Semi-Supervised and Transfer Learning Approaches

- Semi-supervised, transfer, and self-supervised learning enhance NILM performance with limited labeled data [25] [71] [72].

Privacy and Trust Challenges in NILM Deployment

- Privacy concerns are mitigated by federated learning, edge processing, and hybrid approaches; balancing accuracy and privacy remains a challenge [26] [27] [28].

Explainable AI for Load Advice in Energy Platforms

XAI Techniques and Their Application

- SHAP, LIME, Grad-CAM, clustering, and hybrid methods provide transparent, interpretable load advice [29] [30] [73].
- Clustering reduces computational cost while maintaining explanation fidelity [31] [74].

Balancing Explanation Fidelity and Computational Efficiency

- Grad-CAM offers computational efficiency; SHAP provides detailed feature impact insights but is resource-intensive [30] [31].
- Combining multiple XAI methods improves reliability and user trust [75] [76].

User-Tailored Explanations and Trust

- Adaptive, user-centric explanations enhance trust and decision-making; human-centered design is critical [32] [77] [78].

Limitations and Gaps in XAI for Multivariate Time Series

- Curse of dimensionality, high computational costs, and lack of standardized robustness testing limit practical deployment [33] [79] [80].
- Emerging AI architectures (e.g., Transformers) hold promise for addressing these challenges [81].

User-Centric Design Principles for XAI

- Multi-level explanations, workflow integration, and contestability mechanisms enhance effectiveness and trustworthiness [34] [35] [82].

Fair Slot-Based Energy Allocation During Outages

Methodologies and Fairness Criteria

- Slot-based MDP, cooperative game theory (Shapley value, Nash Bargaining), and quota/reputation systems ensure equitable allocation [36] [37] [38].
- Fairness criteria effectiveness depends on building typology and resident profiles [40].

Optimization Frameworks for Fair Allocation

- Multi-objective and stochastic optimization frameworks balance economic benefits and fairness, adapting to uncertainties [39] [83] [84].

Reputation and Quota-Based Management Systems

- Reputation and quota mechanisms prevent exploitation and promote fair utilization in community energy storage [48] [49] [85].

Influence of Building Typologies and Resident Profiles

- Building typologies and demand profiles affect suitability and fairness of allocation models; tailored approaches are necessary [40] [86].

Gaps and Novel Approaches

- Lack of direct comparative studies on fairness criteria; objective proximity analysis needed for model selection [40] [41] [42].

5. Real-world Implications

Benefits and User Impacts

- **Energy savings:** Up to 39% reduction in consumption, improved self-sufficiency, and reduced peak demand [1] [16] [20].
- **User engagement:** Visual dashboards, gamification, and adaptive explanations foster sustained behavior change and empowerment [6] [32].
- **Environmental impact:** Hybrid systems and optimized battery scheduling reduce CO2 emissions and grid reliance [46] [87].
- **Equity and resilience:** Fair allocation frameworks reduce disparities and enhance resilience during outages [37] [38].

Fairness, Privacy, and Trust Issues

- **Fairness:** Game theory, quota/reputation, and adaptive policies ensure equitable resource distribution [2] [48].
- **Privacy:** Federated learning, edge processing, and hybrid approaches mitigate privacy risks in NILM and forecasting [27] [28].
- **Trust:** Explainable AI, user-centered design, and contestability mechanisms build operational transparency and user confidence [32] [78].

Research Gaps and Novel Contributions

- **Prediction interval calibration:** Need for comparative studies between ensemble and Bayesian models [66].
- **Real-time NILM frameworks:** Standardized datasets and reproducible protocols for privacy-preserving deployment [28].
- **XAI for multivariate time series:** Integration of emerging AI architectures and unified frameworks [81].
- **Fairness in allocation:** Objective proximity analysis and comparative studies on fairness criteria [40].

6. Future Research Directions

- **Comparative forecasting studies:** Direct evaluation of ensemble vs Bayesian deep learning for calibrated prediction intervals in apartment energy platforms
- **Standardized NILM frameworks:** Development of real-time, privacy-preserving NILM protocols and datasets for multi-unit settings
- **Advanced XAI integration:** Application of Transformers and Large Language Models within XAI for complex time series forecasting
- **Blockchain-federated learning:** Secure, verifiable energy transaction frameworks for multi-tenant buildings
- **Tailored allocation models:** Objective proximity analysis and adaptive policies for fair energy distribution based on building typologies and resident profiles

7. Conclusion

Summary and Future Directions

Research from 2019–2026 demonstrates significant advances in apartment energy platforms integrating shared solar and batteries. Centralized and hybrid configurations, advanced forecasting, privacy-preserving NILM, explainable AI, and fair slot-based allocation collectively improve operational efficiency, user engagement, fairness, privacy, and trust. Novel contributions include uncertainty-aware optimization for load scheduling and integrated fairness frameworks balancing economic and equity objectives. However, gaps remain in prediction interval calibration, real-time NILM deployment, and XAI for multivariate time series. Future research should focus on comparative forecasting studies, standardized NILM frameworks, advanced XAI integration, blockchain-federated learning, and tailored allocation models to further enhance sustainability, resilience, and equity in apartment energy platforms [1] [12] [20] [29] [36].

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